

Level 2 Augmented Reality System (PokePhy) to Complement Physics Subjects in a Private University in Lima, Peru.

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Abstract

This project, "PokePhy," introduces augmented reality into university-level physics education, employing tools like Unity 3D and Vuforia SDK to create an interactive learning experience. By emphasizing speed, time, and distance, the application offers students two difficulty levels (tutorials and quizzes), facilitating active engagement with the subject matter. While demonstrating the potential of augmented reality in higher education, the project also highlights the need for refined user interfaces and considerations for camera quality and lighting conditions. In doing so, "PokePhy" represents an innovative step towards harnessing technology for enhanced learning experiences at the university level.

Keywords

Augmented reality, education, physics, interactive learning, Unity 3D, Vuforia SDK, gamification.

1. Introduction

Augmented reality technology, as part of immersive technologies [1], has a wide range of applications ranging from healthcare, tourism, military, and aviation; however, in education, it has proven to be a great provider of interactive learning and helps to enhance the learning process experience [2]. Its popularity has been increasing in recent years. It was even found that this popularity was similarly distributed in different countries, mainly in primary to university education in other subjects [3]. This shows the great potential and use of this tool as a learning enhancer, not only in low-level education but also in university education. Due to the constant difficulty of students to find a better way or technique for learning and the increasing, often worrying, use of devices such as cell phones or digital tablets, mainly in universities [4], it is of utmost importance to find new techniques or tools that help students to make educational use of these devices. This can be achieved through augmented reality technology [5].

Specifically, the purpose is, in addition to improving the learning experience through a personalized tool, to improve and enhance some cognitive skills of students. This is because it has been demonstrated that the ability of augmented reality significantly enhances the ability to memorize through gamification [6]. For this reason, it was proposed to develop an application

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with this critical feature, as it is known that the game of people is a driver of skills and abilities, both physical and mental, which translates into greater creativity and imagination [7]. The vast majority of background on the use of augmented reality in education was done in primary education subjects, with the vast majority of these focusing on capturing children's attention, which differs somewhat from the focus of this research; however, the background presented provides valuable information on the use of augmented reality [8] as a factor with a possible significant impact on learning [9] and capable of enhancing students' problem-based learning through promoting a positive attitude and long-term retention of physics-related topics [10].

The tool used in one of the investigations was the free tool Aumentaty [11] to generate augmented reality content quickly. The second research carried out the creation of its FenAR [10] application with the Unity 3D platform and the Vuforia SDK. The first of the latter two is a well-known powerful 3D game engine used in conjunction with augmented reality tools to superimpose the virtual on reality and thus create augmented reality applications or games; at the same time, Vuforia is a raised reality software development kit that uses computer vision to detect one-dimensional images or objects and thus place 2D or 3D models on them [12].

The present research used these two tools to develop the application called PokePhy. This application has a user-friendly interface with highly recognizable character images, demonstrating its video game character. The application consists of two levels: easy and complex, the first one with simple physics problems, while the second one with more complex issues but also closely related to the first mode. These problems were given randomly and tried to complete the classic physics formulas with an invocation of one of the three 3D models. Each model represented a specific dimension, which could be viewed through custom markers.

2. Background

Most of the research related to augmented reality applied in the educational sector is focused on primary education and the development and implementation of applications capable of attracting and improving the attention, and therefore the learning, of children [13, 14]. However, in the case of university students [15], although we did not find much research where augmented reality was applied to teaching, the two precedents show project proposals related to what we want to achieve with PokePhy, in addition to the use of complete tools such as Unity 3D platform and Vuforia SDK. Subsequently, both backgrounds will be described.

The first aimed to develop basic skills in educational technology using mobile devices [16]. This application (shown in Fig. 1) was used to work on different skills in critical thinking, comprehensive thinking, and content analysis. The possibilities that Augmented Reality offers are more significant due to the easy manipulation of objects and teamwork in groups.

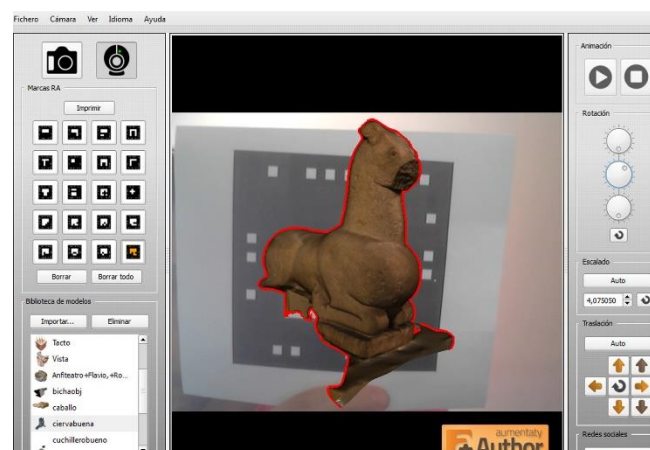


Fig. 1. 3D model in Aumentaty author.

In addition, the methodology used is the project method, which was divided into three phases: The first is about the exploration of images through augmented reality with the devices used. The second phase is the collection of information on the art, culture, and society of the pre-Roman peoples. Finally, the third phase of identifying general characteristics of Iberian art through a comparison of the different works. The subject planned to be taught through the application is Social Sciences in Albacete. The topic touched on in the application is the Iberian art of the same province of Spain.

The results were positive because it is an application that runs on a mobile device. It increases the motivation of the students and stimulates an improvement in learning. Through the figures presented in the application, it awakened the interest and curiosity of the students. In addition, now of learning, they interactively have fun with the classroom dynamics.

The second background [10] (shown Fig. 2) was mainly aimed at evaluating the impact of augmented reality technologies on problem-based learning and physics-related topics in seventh graders in northern Turkey. In addition, a final comparison was made between the results of two experimental groups: one group using augmented reality as a tool for problem-based learning and another group using only augmented reality as a tool for problem-based learning.

Its methodology consisted of a preparation for the students, that is, dividing them into groups with the characteristics mentioned above. Instructions were given on how each group should behave. Then, an integration between augmented reality technology and problem-based learning was performed (at the stages where it was necessary). Finally, an evaluation was conducted to measure the level of knowledge and their attitude toward physics. Their results showed a highly positive impact of augmented reality technology on problem-based learning. Also, they warned that the non-use of this technology in this context leads to a much lower learning performance.

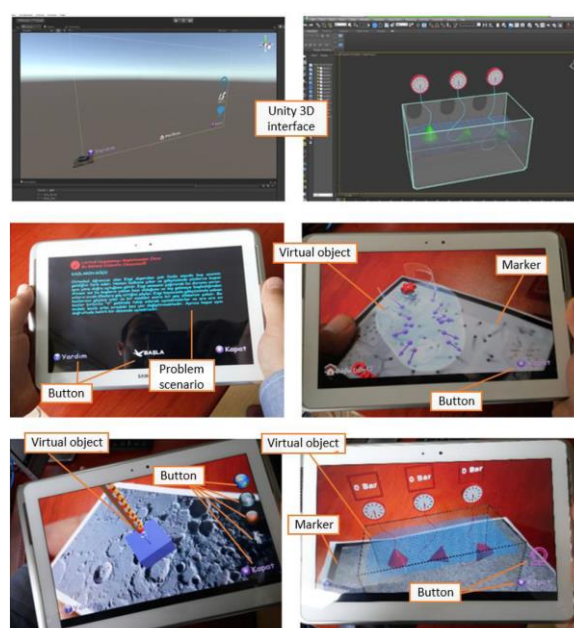


Fig. 2. Design of the FenAR application.

3. Methodology

The main objective of active methodologies is always the same: the participation and cooperation of students in the learning process [17]; however, there are many highly widespread in the educational sector, ranging from problem-based or project-based learning to gamification, which will be used in the research [18]. This type of active methodology has been gaining popularity

lately because it consists of applying elements and principles of games in a more educational or learning environment to achieve greater motivation and engagement for learners [19]. However, these benefits could be better if we consider what gamification entails, mainly the connection between the player and the game. This indicates that the player should find everything he/she needs efficiently to have a positive relationship with the game [20]. Thus, it is proposed to use the waterfall methodology to develop the PokePhy application.

The waterfall methodology (shown in Fig. 3) is a traditional methodology for software development that attempts to simulate the designs of other industries. It divides the entire software project into phases, dependent on the previous stage [21]. In this research, we will follow the points of this methodology, where we will describe how the application was realized through the steps of the methodology diagram extracted from another study [22]. It culminates with a test of the functionality where some university students use the PokePhy application to invoke the 3D elements on the customized cards with a nice theme and playability.

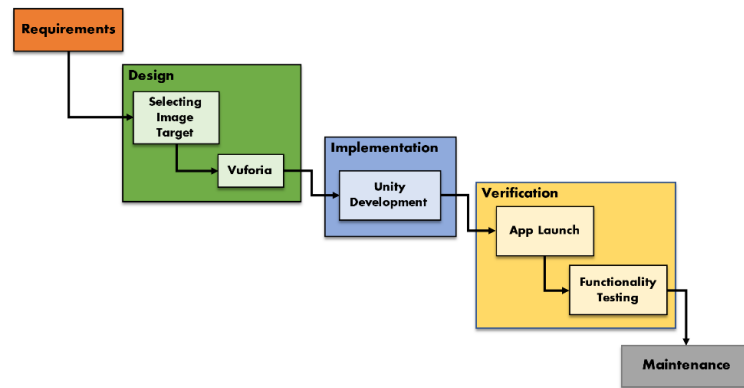


Fig. 3. Diagram of cascade methodology.

3.1. Application Requirements

The list of requirements [23] for the system design is shown in Table 1.

Table 1
List of Requirements

Requirements	Description
1	The application will allow the development of some cognitive skills, such as memory, in young university students.
2	The user will be able to access the application by cell phone.
3	The application can play music, sounds on buttons, and augmented reality interaction.
4	The application will not collect information from its users.
5	The application will not make recordings through the camera.
6	The user can visualize the 3D models through their mobile devices.
7	The application shall have a tutorial section for each of the difficulties.
8	The application shall have two levels of difficulty.
9	The interface design of the application must be following the theme of the game.
10	The application shall be compatible with Android devices.

3.2. Selection of Image Target and Multimedia Resources

First, we must select the images that will serve as markers to be scanned by the app and displayed by it. For the implementation of all this, we will use Vuforia Engine and Unity. We will use images with the first letter of every word that every formula that we will use has, such as Velocity, Time, and Distance. Then, we must convert them to 24-bit format, which Vuforia allows using images. It will be used as the augmented reality graphics engine software to enable us to create the application using this technology. We will start adding each image we have to a database (shown in Fig. 4) in Vuforia, and it will give us a score that defines the quality of the picture.

Type: Device

Targets (3)

Add Target

Download Database (All)

☐ Target Name	Type	Rating ^①	Status [▼]	Date Modified
☐  velFinal	Image	★★★★★	Active	Jul 02, 2023 10:08
☐  timeFinal	Image	★★★★★	Active	Jul 02, 2023 10:07
☐  distFinal	Image	★★★★★	Active	Jul 02, 2023 10:07

Fig. 4. Database of images.

Subsequently, the database will be downloaded as an installation package (shown in Fig. 5) to import it to Unity. Finally, we will add from Image Target and assign it to the images that Vuforia allows us to use. We must have Vuforia's AR Camera enabled to utilize the free version provided.

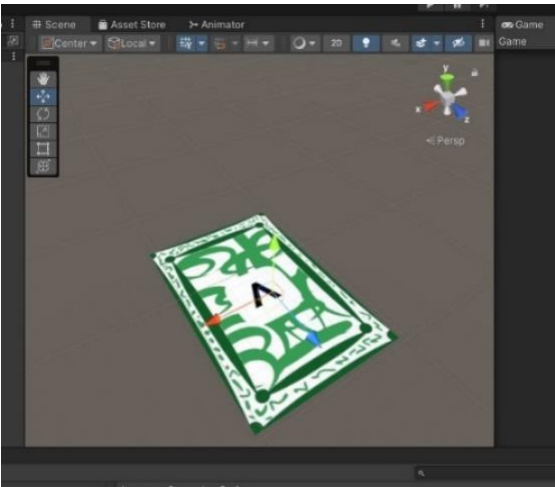


Fig. 5. Insert Database of images to Unity.

3.3. Application development in Unity

We will commence by creating the user interface using the Canva tool provided by Unity, and we will set the screen size to 2196 x 1080. Furthermore, we will integrate the Vuforia augmented reality camera, which will enable us to ensure the proper interaction between the image and the 3D model. Concerning the user interface (shown in Fig. 6), we have designed buttons to facilitate communication between interfaces, allowing the game options to be executed within the augmented reality environment.



Fig. 6. Development of the application interface in Unity.

An image target of the completed puzzle has been used so that, after solving it, the visualization of a 3D model starts. This model has been animated on the Mixamo page, with many movements stored in a mocap database in BVH format. In this way, the 3D model can be customized. With Mixamo, any character can be animated; then, the model is downloaded and linked to the image (shown in Fig. 7).

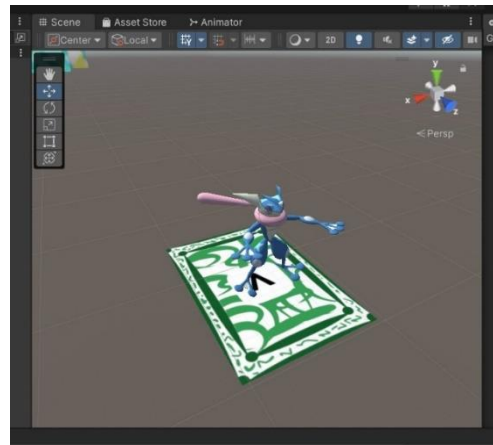


Fig. 7. Implement the 3D model and videos to the image targets.

3.4. Functional description

Regarding functionality, the project has been designed intuitively, with each part of the application serving a purpose in teaching this Physics topic, which involves speed, time, and distance. Regardless of the user's characteristics, they can learn about the subject by following a sequence and making decisions as it suits them.

The first that users can do when they run the application is access the tutorials that are part of this Physics learning experience. These tutorials include formulas encompassing previously mentioned concepts, ranging from foundational knowledge to slightly more complex equations. Following this, they can choose one of the two available game modes: easy and hard. Both game modes open an augmented reality experience where they can use external cards, each containing one of the concepts, to respond to the presented equations, thus reinforcing their understanding of the formulas.

Users should start in accessible mode and wish to increase the difficulty level. They can switch to hard mode by using a button that will be available. In both modes, they can view their score once they complete all the levels available for that game mode.

3.5. User Interface Description

Now, we will describe and present an overview of the application's interface (shown in Fig. 8). The interface has been designed to be colorful and in line with the concepts it aims to teach, as it is oriented towards learning physics. Therefore, the application's background consists of an animated image in green and blue colors, and there is also a looping melody. Upon launching it, you can see the presentation of the program used for its creation. In this case, it is Unity. Following this, the application will request the necessary permissions to access the device's camera each time it is run.

Once permissions are granted, you can see the start of the main interface, which includes the application's name, an image of a Poke ball, and a Pokémon representing the game's central theme. The user should click on the image of the Poke ball to access the main menu of the application, where four buttons of different colors can be seen, along with an animation of more Pokémon. These four buttons consist of two game modes and two tutorials.

In Tutorial 1, you can see the types of Pokémon corresponding to each concept being studied, such as distance, speed, and time. It is even possible for the user to click on each Pokémon to hear its characteristic associated sound, which will also be heard during augmented reality. The user can return to the main menu using the back arrow button. Tutorial 2 presents a more advanced level of these concepts, including various formulas that use them and their results. This page also includes a back arrow button that allows returning to the main menu.

Moving on to the game modes, the user can choose between the easy manner and the problematic mode. In an accessible way, the application will open the device's camera, and the user will be able to see the score increasing or decreasing in the corners, as well as the number of levels completed, up to six. Additionally, the application will generate a fundamental problem using the previously mentioned concepts. The user should focus on cards representing distance, speed, and time as a response to the equation presented. This will also display the corresponding Pokémon for each term as an augmented reality image, aiding in better memorizing these formulas. Once the six rounds are completed, the game will show a game over, indicating it's finished.

In the challenging game mode, the user will perform the same action but using equations (seen in Tutorial 2) with a higher difficulty level. Similarly, once seven rounds are completed, the game will end.

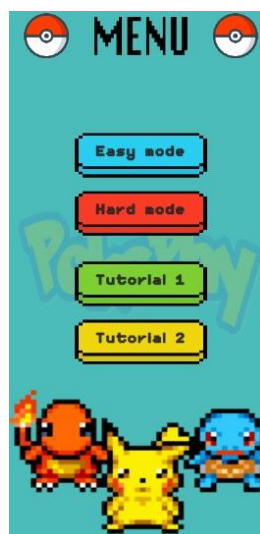


Fig. 8. Start of the PokePhy application, own creation.

3.6. Functionality tests.

In this case, a functional test was carried out on a sample of 50 university students in the third and fourth cycles of Physics I. The function of the application was explained to the participants, and they were divided into groups. They were given a link to download the app on their mobile device. All materials used in the test were virtual, both the application and the image cards where the 3D models would be projected.

The students worked in pairs (shown in Fig. 9); one person opened the application, and the other accessed the image on the card. It is important to note that due to operating system security limitations, the app can only be installed on Android devices, regardless of phone model.

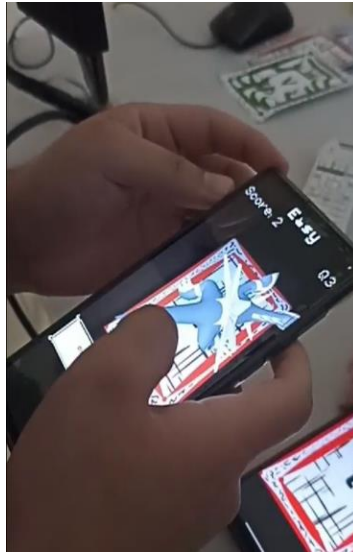


Fig. 9. Students solve the problems in accessible mode.

Testing with the application installed (shown in Fig. 10) was carried out as follows: First, when students launched the app, they were presented with a fully graphical interface with four options, including tutorials and quizzes. If you choose the guide, you will receive detailed instructions on how to use the app. If you select an examination, you can complete the task and present the card you think is the correct answer.

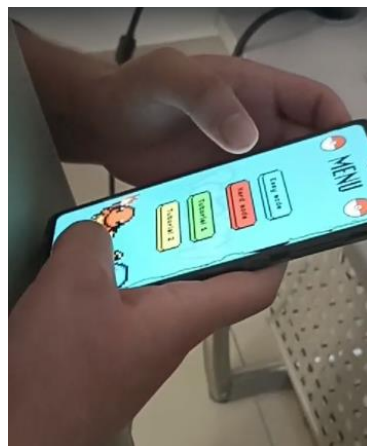


Fig. 10. Students using Pokephy's interface.

During testing (shown in Fig. 11), we discovered that camera quality and lighting conditions can affect the time for 3D models to appear. However, every effort has been made

to minimize this problem, resulting in the model occurring reliably within seconds despite some difficulties.



Fig. 11. Despite showing the correct answer, the game finished because of the camera.

Additionally, the interface turned out to be less intuitive for users, making the game sometimes confusing. We found some minor design flaws that must be fixed to improve the user experience.

4. Conclusions

In conclusion, integrating augmented reality into education, as evidenced by the "PokePhy" project, represents a promising avenue for enhancing the learning experience, particularly in higher education. While most augmented reality research has been concentrated in primary education, this project's alignment with the established methodologies and its focus on teaching physics to university students opens exciting possibilities. Utilizing tools like Unity 3D and Vuforia SDK demonstrates a commitment to harnessing advanced technology for educational benefits.

However, the project also highlights some important considerations. The need for improved user interface design and the impact of camera quality and lighting conditions on the augmented reality experience underscores the importance of refining the application's usability. These challenges can be overcome through iterative development and close attention to user feedback, ultimately paving the way for augmented reality to play a more prominent role in enriching the educational journey for university students. As technology continues to evolve, the "PokePhy" project exemplifies the innovative spirit that seeks to unlock the vast potential of augmented reality in higher education, offering a promising path forward for educators and learners alike.

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